

Egocentric Video Question Answering: Temporal Segment Retrieval and Natural Language Response Generation

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Abstract

Egocentric videos capture daily life from a first-person perspective and are typically long and unstructured, which makes finding relevant information challenging. We address the Natural Language Queries (NLQ) task on the Ego4D benchmark, aiming to identify video segments that answer a user’s natural language question. We train and compare VSLBase and VSLNet using pre-extracted Omnivore and EgoVLP features, evaluating their performance on the validation set against the official SlowFast baseline. We also explore a VSLNet variant with separate encoders for video and text, assessing its impact on localization accuracy. Building on these results, we propose an extension that generates textual answers: the pipeline retrieves the relevant interval using the outputs of the top-performing model and then applies Video-LLaVA to produce an answer. To evaluate the full pipeline, we manually annotate 50 query-segment pairs with ground-truth answers for quantitative and qualitative assessment. This work bridges segment retrieval and natural language understanding in egocentric videos, highlighting the potential of integrating temporal localization with video question answering. The full implementation is available at https://github.com/auroraji2906/Ego4D_EpisodicMemory_Group19.

1. Introduction

Egocentric video understanding provides a first-person perspective on human activities, but the long and unstructured nature of such videos makes it challenging to identify segments relevant to specific queries. The Ego4D dataset [6] addresses this with the Natural Language Queries (NLQ) benchmark, which requires temporally localizing

video segments that answer free-form natural language queries. Unlike traditional action recognition, this problem requires fine-grained alignment between video and language, as well as robust reasoning over long temporal contexts. Recent methods, such as VSLBase and VSLNet, tackle NLQ by jointly encoding video and text features; VSLNet further incorporates a query-guided highlighting mechanism to focus on relevant frames.

Building on these methods, our work investigates the role of domain-specific pretraining, architectural design, and cross-modal alignment in egocentric NLQ. We evaluate two visual backbones: EgoVLP, pretrained on egocentric video-text pairs, and Omnivore, a general-purpose model trained across multiple modalities. Textual queries are encoded using frozen BERT embeddings. We experiment with both VSLBase and VSLNet under shared and non-shared encoder configurations to assess their different impact on temporal retrieval.

While temporal localization identifies relevant video segments, understanding and summarizing their content requires further reasoning. To address this, we extend our pipeline to egocentric VideoQA, where retrieved segments are fed into Video-LLaVA, a vision-language model capable of generating natural language answers that summarize the relevant visual content in the context of the query.

In our experiments combining VSLNet with EgoVLP features achieves the best performance among the tested models. For VideoQA, moreover, Video-LLaVA demonstrates strong semantic understanding but struggles with precise temporal reasoning and numerical accuracy.

2. Related Work

Research in egocentric vision has advanced significantly with the release of large-scale datasets and specialized architectures. Among these, the Ego4D dataset [6] stands out

for its scale and diversity offering 3,670 hours of unstructured egocentric video captured by 931 participants across 74 locations worldwide. It covers a wide range of everyday scenarios and supports multiple benchmarks, including Natural Language Queries (NLQ), which focus on retrieving temporally grounded answers to user-specified textual prompts. Other domain-specific datasets, like EPIC-Kitchens-100 [2] focus on kitchen activities and have driven progress in action recognition and temporal segmentation, despite their smaller scale (around 100 hours of annotated footage).

To address the NLQ task, several models have been proposed based on span prediction strategies. For instance, TALL [4] directly predicts start and end times of video segments matching a natural language query. VSLNet [9] extends this approach with a query-guided highlighting (QGH) mechanism that emphasizes relevant video segments, while its simplified variant, VSLBase, removes this component but maintains competitive performance. Alternative methods, such as 2D-TAN [10], explicitly model temporal adjacency between video segments to improve alignment with textual queries.

While most works have focused on refining model architectures, the choice of visual backbone—i.e., the feature extractor used to encode raw video frames—has received comparatively less attention, despite its significant impact on overall performance. In the official Ego4D NLQ benchmark, SlowFast [3] is adopted as the default visual backbone. However, recent research has explored alternative backbones. General-purpose models such as Omnivore [5] can process multiple visual modalities within a single network, offering broad versatility across tasks. On the other hand, EgoVLP [8], pretrained on egocentric video–text pairs, aligns visual and textual embeddings via contrastive learning and captures fine-grained semantic details, making it particularly effective for NLQ. Considering these complementary characteristics, we evaluate both backbones to study their impact on temporal localization performance.

Beyond temporal localization, recent research has explored generative vision-language models that produce natural language answers directly from video input. A representative example is Video-LLaVA [7], which combines pretrained multimodal encoders (e.g., LanguageBind [11]) with large language models to generate contextualized responses grounded in visual input. These approaches demonstrate promising capabilities in cross-modal reasoning and open-ended question answering, but often lack precise temporal grounding necessary for complex egocentric queries.

Our pipeline aims to combine robust retrieval models (VSLBase and VSLNet) with generative ones (Video-LLaVA) to generate fully articulated, context-aware an-

swers grounded in retrieved video content. This integrated approach bridges the gap between temporal localization and natural language generation, enabling richer egocentric video understanding.

3. Dataset

The Ego4D dataset comprises over 3,670 hours of egocentric video collected from 931 participants across 74 locations worldwide, covering a wide range of daily activities, including household tasks, leisure, work, navigation, and social interactions, recorded using head-mounted cameras to provide a first-person perspective. Additional metadata, including gaze and audio, is available but not used in our experiments.

In this work, we focus on the Ego4D Natural Language Queries (NLQ) benchmark version 1 [6], which contains approximately 19,000 annotated queries from around 227 hours of untrimmed video. Each video is segmented into clips, and each query annotation includes:

- Temporal boundaries: start and end timestamps, relative to both the clip and the full video.
- Natural language queries: textual descriptions of the user’s question about a video segment.
- Template information: each query is assigned to one of 13 templates, together with slots and verbs that specify the relevant entities and actions. As shown in Figure 1, template distribution reveals a strong bias toward object-centric questions (e.g., locating, counting, or tracking objects), while queries involving people or social interactions are comparatively rare.
- Raw tags: summarizing query, slots, and verbs, useful for normalizing linguistic information across annotations.

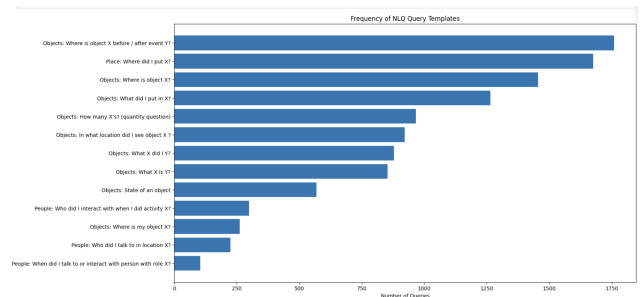


Figure 1. Frequency distribution of NLQ query templates, ordered by decreasing frequency (most frequent at the top)

The clips have an average duration of approximately 523 seconds, with most around 480 seconds, as shown in Figure

2. Durations ranges between 207 and 1,200 seconds, which increases the complexity of temporal localization. Each clip is associated with multiple natural language queries, where the target temporal segments are usually short compared to the full video: most occupy less than 20% of the clip duration and very few exceed 30%. In particular, approximately 9% of queries are extremely short (four frames or fewer) and are removed before training to ensure stable learning.

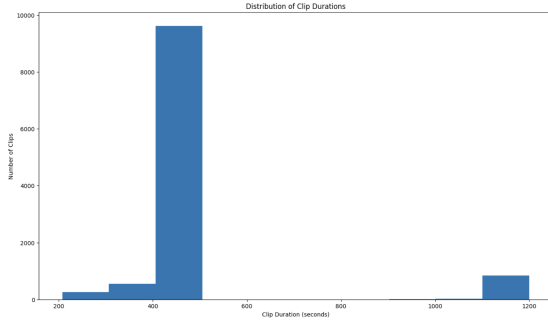


Figure 2. Distribution of clip durations.

From a semantic perspective, the dataset is strongly biased toward domestic and practical scenarios, such as cooking, cleaning, and maintenance activities. Queries set in public or technical environments (e.g., shopping, mechanical repair) are less represented, as illustrated in Figure 3, which can limit the generalization capability of models outside household domains.

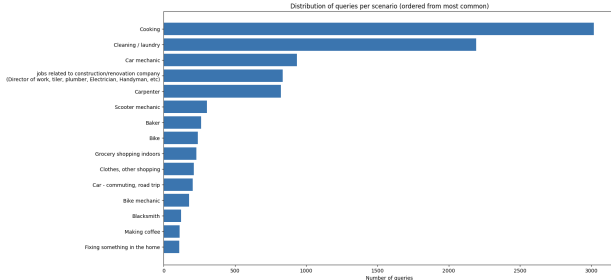


Figure 3. Top 15 scenarios by number of queries, ordered from most to least frequent (most frequent at the top).

Overall, NLQ represents a challenging benchmark, requiring models to localize short and sparse events within long untrimmed videos while dealing with biased scenario and query-type distributions.

4. Methodology

In this section, we describe the methodology employed to address the Natural Language Video Localization (NLVL) task. We first present the pre-extracted visual and textual features used as input to our models. Next, we detail

the temporal localization architectures, including VSLBase, VSLNet, and a non-shared encoder variant. Finally, we outline how the predicted temporal intervals are used to extract the corresponding video content and generate textual answers, following the workflow illustrated in Figure 5.

4.1. Pre-extracted Features

To reduce the computational cost of training on long egocentric videos, we rely on pre-extracted features from pre-trained backbones for both the visual and textual modalities, allowing models to focus on learning the alignment between video and language representations.

For the visual modality, we use features from EgoVLP and Omnivore backbones. EgoVLP [8] is pre-trained on large-scale egocentric video datasets, enabling it to capture hand-object interactions and the temporal regularities typical of first-person perspectives.

Omnivore [5], in contrast, is trained jointly on heterogeneous datasets including ImageNet, Kinetics, and ScanNet, enabling modality-agnostic representation learning across images, videos, and 3D inputs. Its inclusion provides robust and modality-agnostic representations that help the model generalize across heterogeneous visual inputs. The previous two features are extracted at fixed temporal intervals (32-frame windows with stride 16).

For comparison with the official Ego4D NLQ benchmark, we also report results using the SlowFast backbone [3], a spatiotemporal convolutional network designed to capture motion dynamics at multiple temporal resolutions. Unlike EgoVLP and Omnivore, SlowFast is trained primarily on general video datasets and does not leverage egocentric video-specific pretraining.

For textual modality, we use contextualized embeddings from BERT, which provide robust linguistic representations of the natural language queries. These embeddings are also frozen ensuring that subsequent components in the pipeline concentrate on multimodal alignment. By comparing localization models trained on EgoVLP and Omnivore visual features with the SlowFast baseline, we can assess the impact of domain-specific and general-purpose backbones on temporal grounding performance.

The architectural details of these models are described next.

4.2. Temporal Localization Models

We adopt a span-based framework predicting the start and end boundaries of the answer segment corresponding to a natural language query. Our approach is based on two closely related architectures: VSLBase and VSLNet, which extends VSLBase with a Query-Guided Highlighting (QGH) module (see Figure 4).

Both models first map visual and textual features into a shared feature space. Video features (pre-extracted from

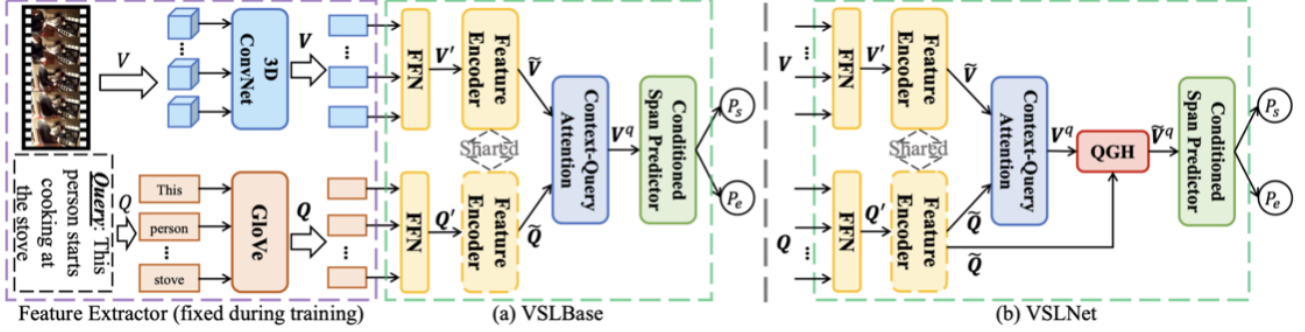


Figure 4. Architecture of the VSLBase and VSLNet models.

backbones such as EgoVLP or Omnivore) and textual embeddings (from BERT) are linearly projected to the same dimensionality. The projected video features are treated analogously to textual passages, enabling the model to apply sequence modeling techniques for temporal localization. These features are then processed through a shared encoder that captures local temporal dependencies via stacked convolutions and long-range interactions through multi-head self-attention. Feed-forward layers, combined with residual connections and layer normalization, refine the representations and stabilize training.

After encoding, context-query attention mechanism aligns video and text modalities, highlighting frames most relevant to the query. A conditioned span predictor, composed of stacked unidirectional LSTMs and feed-forward layers, then produces probability distributions over the start and end positions. The end boundary prediction is conditioned on the hidden state of the start boundary, capturing temporal dependencies between the two boundaries and ensuring coherent segment predictions.

Training in VSLBase is guided by a cross-entropy loss applied to the predicted boundaries (see Zhang et al., 2020 for details).

VSLNet extends this design with a Query-Guided Highlighting (QGH) module, which introduces an auxiliary foreground/background classifier over the encoded video tokens. The QGH module assigns a highlighting score to each frame, biasing the model to emphasize regions of the video more likely to contain the answer segment. These scores are used to re-weight the visual features before boundary prediction. As in VSLBase, the span predictor is conditioned so that the end boundary leverages information from the start boundary. Training minimizes a combined loss: the span prediction loss plus the foreground-background classification loss from QGH (see Zhang et al., 2020). This extension allows VSLNet to better capture subtle temporal differences between frames, improving localization accuracy, particularly for long and unstructured egocentric videos.

4.3. VSLNet Variant with Non-Shared Encoders

To further investigate the role of modality-specific encoding, we developed a variant of VSLNet in which the video and text features are processed by independent encoders rather than a shared one. The overall architecture follows the original design, but two separate instances of the Feature Encoder are instantiated: one dedicated to visual features and the other to textual embeddings. Both encoders share the same hyperparameter configuration (e.g., number of layers, attention heads, kernel size), but their weights are initialized independently, ensuring that the representations evolve without cross-modal parameter sharing.

Their outputs are combined using VSLNet’s context-query attention, feature concatenation, highlighting, and conditioned span prediction mechanisms. Evaluating this variant alongside the base models with different visual feature sources enables us to assess whether non-shared encoders can enhance temporal localization performance.

4.4. Textual Answer Generation from Predicted Segments

Building upon the temporal localization models, we extend the pipeline to generate textual answers for egocentric video queries. While NLQ models provide start and end timestamps for relevant video segments, they do not directly yield natural language answers. To obtain the answer, a user would need to watch the retrieved segment. This limitation is particularly relevant in egocentric videos, which are long, unstructured, and visually complex. To address this, we extend temporal localization, by employing Video-LLaVA to produce natural language answers. The full pipeline is shown in the Figure 5.

Firstly, we use the predictions of the temporal localization models described in Section 4.2. We select predictions from the best-performing checkpoint on the validation set, measured by Recall@1 and mIoU@0.3. To ensure high-quality input for answer generation, predicted segments are ranked according to their temporal Intersection-over-Union

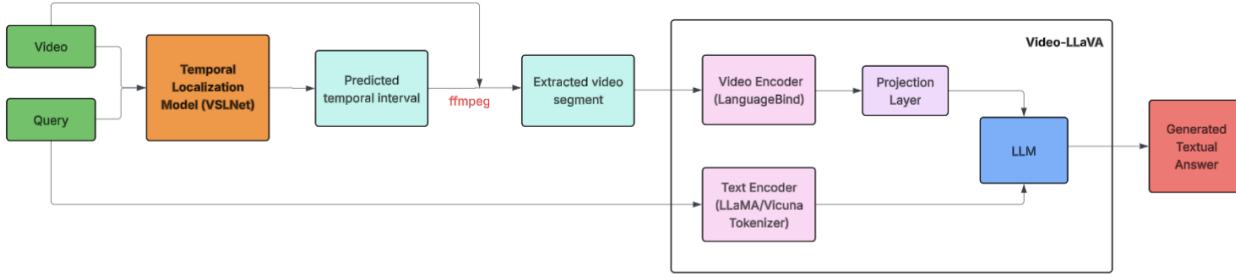


Figure 5. Full pipeline for producing natural language answers from temporally localized segments.

(IoU) with ground-truth annotations, and the top 50 queries are retained. These segments are then extracted from the original video recordings using standard video processing tools, such as *ffmpeg*, ensuring that the input to the next stage contains only the relevant visual content. Each segment is paired with a manually annotated textual answer, formulated as a concise declarative sentence capturing key visual content and context, that serves as references for evaluating generated answers.

For answer generation, we employ Video-LLaVA, a multimodal video-language model extending the LLaVA framework to temporal video understanding.

Video-LLaVA encodes video frames using the pre-trained LanguageBind [11] encoder and tokenizes the textual query via the LLaMA/Vicuna tokenizer. A projection layer then maps the video embeddings into the LLM’s embedding space, enabling joint reasoning across visual and textual modalities. For each video segment, a subset of frames, uniformly sampled (in our setup, it is used 8 frames per segment to reduce computational cost), is extracted and processed together with the input query.

The LLM generates a fluent, semantically grounded textual answer that reflects both the visual content of the segment and the query context. In addition to the standard model pipeline, we apply a post-processing step to clean the generated answers, removing redundant punctuation, special characters, and other artifacts, ensuring higher readability, consistency, and fidelity to human-annotated references. Generated answers then are evaluated using BLEU-1 and BLEU-4, ROUGE-L, and METEOR [1], which together assess lexical precision, sequence similarity, and semantic alignment with human-annotated answers.

Overall, this two-stage pipeline demonstrates how temporal localization can serve as a critical preprocessing step for video question answering, reducing computational complexity while enabling the generation of concise and contextually grounded textual answers from long and unstructured egocentric videos.

5. Experiments and Results

In this section, we present a comprehensive evaluation of our temporal localization models on the Ego4D-NLQ validation set, analyzing the impact of architecture choice (VSLBase vs. VSLNet), visual feature backbone (EgoVLP vs. Omnivore), and encoder design (shared vs. non-shared). Furthermore, we extend the evaluation to assess the quality of natural language answers generated by the Video-LLaVA model based on the top predicted segments.

5.1. Experimental Setup

All models were trained for 20 epochs with a learning rate of 0.002 using the AdamW optimizer and a batch size of 32. We set the hidden dimension to 128 and the maximum sequence length (i.e., the maximum number of tokens considered per query) to 128. This setup was chosen to balance computational efficiency and stable convergence under limited resources, contrasting with the original baseline’s longer training schedule (200 epochs, LR=0.0001).

The evaluation for the temporal localization models is performed using:

- **IoU (Intersection over Union):** measures the overlap between the predicted interval P and the ground-truth interval G , formally defined as:

$$IoU = \frac{|P \cap G|}{|P \cup G|},$$

where $|P \cap G|$ represents the duration of the intersection between predicted and ground-truth intervals and $|P \cup G|$ represents the duration of their union.

- **Recall@k:** reports the percentage of queries for which at least one of the top-k predicted intervals has an IoU exceeding a specified threshold.

In our experiments, results are reported using Recall@1 and Recall@3 at IoU thresholds of 0.3 and 0.5.

5.2. Performance on NLQ task

We conduct a detailed ablation study focusing on three key factors, summarizing the result in Table 1:

- architecture choice (VSLBase vs VSLNet),
- visual backbone (EgoVLP vs Omnivore vs SlowFast),
- shared vs non-shared encoders on VSLNet

Model	Features	IoU=0.3 (%)		IoU=0.5 (%)	
		r@1	r@3	r@1	r@3
<i>Shared Encoders</i>					
VSLBase	EgoVLP	9.06	13.96	5.76	9.16
VSLBase	Omnivore	6.48	10.33	3.82	5.99
VSLNet	EgoVLP	9.60	15.07	5.68	9.76
VSLNet	Omnivore	7.15	11.10	4.08	6.81
VSLNet (Baseline)	SlowFast	5.45	10.74	3.12	6.63
<i>Non-shared Encoders</i>					
VSLNet	EgoVLP	9.73	15.18	5.55	9.73
VSLNet	Omnivore	7.28	11.49	4.21	7.07

Table 1. Results on the validation set for Ego4D NLQ

Architecture. VSLNet generally outperforms VSLBase across visual backbones. For instance, when using EgoVLP features, VSLNet achieves a Recall@1 of 9.60% at IoU=0.3 and 5.68% at IoU=0.5, compared to 9.06% and 5.76% for VSLBase. A similar trend is observed with Omnivore features (7.15% vs. 6.48% at IoU=0.3, Recall@1). This improvement aligns with expectations from the Query-Guided Highlighting (QGH) module, which helps the model focus on frames most relevant to the query, disambiguating subtle temporal cues in egocentric videos and improving span prediction accuracy.

Feature backbones. As shown in the Table 1, EgoVLP consistently outperforms Omnivore, and the following analysis highlights the reasons behind this gap. EgoVLP’s domain-specific pretraining on egocentric video-text pairs allows finer alignment between query semantics and video segments, resulting in higher scores. Omnivore, while effective as a general-purpose model, lacks task-specific alignment for egocentric scenarios, leading to comparatively lower scores. The SlowFast baseline, designed for spatio-temporal video recognition, exhibits the weakest result, likely due to the lack of domain-specific adaptation.

Impact of Non-Shared Encoder Variant. By introducing non-shared encoders, we achieve the best-performing model: VSLNet with EgoVLP features and non-shared encoders, achieving Recall@1 of 9.73% and 5.55% at IoU=0.3 and 0.5, respectively, and Recall@3 of 15.18% and 9.73% at IoU=0.3 and 0.5, respectively. This configuration

allows the video and text modalities to independently map their embeddings into a common space before fusion, supporting modality-specific specialization: the video encoder focuses on visual patterns, while the text encoder captures semantic structure. However, in our setting, this design provides only marginal improvements over shared encoders (+0.13% r@1 at IoU=0.3 with EgoVLP). This limited gain is likely due to the use of frozen, pre-extracted features, where the shared encoder already effectively fuses the available embeddings. The advantage of non-shared encoders may become more significant in an end-to-end training scenario where both encoders are fine-tuned jointly.

5.3. Textual Answer Generation

The second stage of the full pipeline extends temporal localization to video question answering. We evaluated the textual answer generated by Video-LLaVA using the top-50 predicted segments obtained from the best VSLNet checkpoint with EgoVLP features. Then the generated natural language answers are compared against manually annotated ground-truth responses.

5.3.1 Evaluation Metrics and results

Generated answers are evaluated using three complementary metrics:

- BLEU: assess the degree of n-gram overlap between the generated and reference answers. BLEU-1 focuses on unigram matches, capturing individual words accurately, while BLEU-4 evaluates sequences of up to four words. Both metrics apply a brevity penalty when the generated answer is shorter than the reference, ensuring that short outputs are not artificially rewarded.
- ROUGE-L: assesses content coverage via the Longest Common Subsequence, combining precision and recall to account for both the proportion of reference content captured and the preservation of word order.
- METEOR: complements these metrics by aligning words through exact matches, stemming, and synonyms, while penalizing disordered alignments; thus, it captures both lexical overlap and semantic similarity, accounting for cases in which different terms express the same concept.

Together, these metrics allow for a nuanced assessment of model performance. Counting or timing questions are best evaluated with BLEU-1, which captures correct individual words regardless of order, while phrase-level or ordering-sensitive questions benefit from BLEU-4, which accounts for sequences of up to four words and the correct word order. ROUGE-L and METEOR are more suitable for object

recognition, spatial reasoning, or interaction questions, as they better capture semantic fidelity and partial correctness..

On the 50 query-segment pairs, the model achieved average scores of BLEU-1 = 0.5571, BLEU-4 = 0.3854, ROUGE-L = 0.6346, and METEOR = 0.6802. These results indicate that the generated answers generally capture much of the meaning and semantic structure of the reference answers, though minor lexical differences, counting discrepancies, or inaccuracies in precise details may still occur.

5.3.2 Qualitative results

A closer inspection of the 50 query-answer pairs reveals specific scenarios where the Video-LLaVA either succeeds or fails. In this section, we present a subset of representative examples that illustrate how the model interprets visual and temporal context, highlighting both its strengths and its limitations. Additional examples are provided at this [link](#).



Figure 6. **Query:** What tool did I use on the machine first?

GT answer: The tool you used first on the machine was an electric screwdriver.

Generated answer: The tool you used first on the machine was a screwdriver.

The model shows consistent ability to capture the overall context and to answer correctly when the scene is visually clear and the action is straightforward. For example, in the query “What tool did I use on the machine first?” (Figure 6), Video-LLaVA correctly predicted “screwdriver,” which is considered correct even though it misses the detail “electric” in the ground truth. This case illustrates the model’s capacity to combine action recognition with object identification, effectively grounding its response in salient visual cues. Such tasks, where the object is visually distinct and the temporal sequence is unambiguous, represent scenarios in which the model performs reliably.

Despite these strengths, the model struggles in more complex scenarios, particularly when objects visually resemble one another. Even when objects are clearly visible to humans, Video-LLaVA may still confuse visually distinct items.

In Figure 7, for example, it confuses a shelf with a fridge. This occurs because the model operates on embeddings extracted from frames, where objects with similar color or texture can be close in feature space. Additional factors, like unusual viewpoints, partial occlusions, varying illumination, cluttered backgrounds, and the influence of the text query during multimodal fusion can bias the model toward semantically plausible but incorrect answer.



Figure 7. **Query:** Where was the plate before I picked it up?

GT answer: The plate was in a drawer before you took it.

Generated answer: The plate was in the refrigerator before you picked it up. (from the refrigerator shelf).

Another major limitation lies in queries requiring exact counting or numerical reasoning. While the model can generally identify which objects are relevant, it fails in providing exact quantities.

As illustrated in Figure 8, Video-LLaVA overestimates the number of doughs onto which egg was applied. Although it correctly identifies the relevant objects, it struggles to enumerate them accurately. This failure can be at-



Figure 8. **Query:** How many doughs did i apply egg on?

GT answer: You applied egg on ten doughs.

Generated answer: I applied egg on 12 doughs. (croissants).

tributed to several factors. Rapid hand movements may partially occlude some dough pieces, while their visual similarity increases ambiguity. Additionally, the model tends to process each frame or short sequence independently, which limits its ability to perform object tracking or aggregate in-

formation over time. As a result, it can miss objects or miscount actions that unfold over multiple moments. The fact that we chose in our setting to process only 8 frames (see Section 4.4) per segment further limits visual coverage, increasing the likelihood of errors in object recognition or action tracking.

6. Discussion and findings

In our experiments VSLNet consistently outperforms VSLBase, confirming the effectiveness of the Query-Guided Highlighting mechanism for focusing on relevant frames in long egocentric videos. Among visual backbones, EgoVLP achieves the best performance, highlighting the importance of domain-specific pretraining on egocentric video-text pairs for precise temporal alignment. General-purpose models like Omnivore and SlowFast show lower accuracy, emphasizing the need for task-specific representations. The non-shared encoder variant provides only marginal improvements, suggesting that frozen pre-extracted features already capture sufficient modality-specific information for temporal localization under our computational constraints.

While the pipeline is able to retrieve relevant video segments and produce semantically meaningful answers, performance is limited in several scenarios. Challenges arise with visually similar objects, occlusions, brief actions, or tasks requiring exact counting. Furthermore, limited frame sampling and a relatively short training schedule, due to computational constraints, likely affected performance. Extended training or denser frame coverage could improve both temporal localization and answer quality, particularly in complex or ambiguous scenes. Overall, our work demonstrates that combining strong temporal localization with multimodal reasoning provides a promising approach for bridging video retrieval and natural language understanding in egocentric videos.

Future improvements could include end-to-end finetuning, denser temporal coverage, and explicit mechanisms for counting or tracking objects and actions within the retrieved segments, which would further enhance the accuracy and reliability of the generated answers.

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